

Ansh Bhatnagar

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EDUCATION

University of California, San Diego

B.S. in Data Science and Cognitive Science

- Minors in Mathematics-Computer Science and Supply Chain Management; GPA: 3.90

La Jolla, CA

Sept 2023 - June 2027

EXPERIENCE

SDE Intern

June 2025 – September 2025

Amazon Web Services

San Diego, California

- Designed and implemented an unsupervised ML-based anomaly detection service for AWS EC2 deployments, enabling automated detection of abnormal behavior at scale and accelerating rollback decisions during hardware rollouts.
- Built and productionized the service using AWS CDK, SageMaker, Lambda, and S3, integrating it into existing deployment protocols where it has been adopted by full-time engineers.
- Gained on-call exposure to resolve real-time client errors under incident scenarios.

Data Science Intern

October 2024 – June 2025

Hey Social Good

La Jolla, California

- Worked with the CEO to operationalize machine learning-based classification of sustainability data, converting 10,000+ subjective data points into consistent scores that informed internal sustainability evaluations and dash-boarding results.

Research Assistant

April 2025 – Present

Chiba Lab, UC San Diego

La Jolla, California

- DataOrganisms: Built an end-to-end full-stack app and real-time IoT data pipeline supporting live classroom analytics across 30+ embedded devices for research-grade data collection and a forthcoming publication.
- AI Classroom: Engineered a computer vision and signal processing pipeline over biometric and video data to process large-scale student learning streams and extract actionable insights for learning science research.

Instructional Assistant

January 2025 – Present

Hahcioğlu Data Science Institute, UC San Diego

La Jolla, California

- Supporting 200+ students in Database Management Systems and Theoretical Foundations of DS by maintaining the course website, hosting office hours, and developing SQL, Python, and ML demos to enhance student learning.

LEADERSHIP

President | Data Science Student Society (DS3)

- Lead 600+ members; launched Quarterly Projects, Consulting, Podcast, LeetCode sessions; & expanded collaborations.
- Primary organizer of [DataHacks 2026](#) (400+ participants); raised \$75K+ in sponsorship.

Captain | AWS Cloud Club at UC San Diego

- Founded and led cloud/ML workshops and industry partnerships focused on AWS infrastructure and GenAI.

President | Google Developer Group (GDG) on Campus at UC San Diego

- Founded chapter; hosted technical workshops and speaker events focused on Google's Build With AI & GCP.

PROJECTS

Fire-Ready Forests | *Python, Machine Learning, PyTorch*

- Led end-to-end ML modeling and data engineering for ecological tree classification, achieving 89.6% PFT accuracy and ~99% R^2 CBH/CR imputation across 500K+ TLS records.
- Won 2nd place nationally in the Data Platform Challenge for model performance and technical implementation.

Comparative Analysis of Stable Diffusion & cDC-GANs | *Python, PyTorch, GANs*

- Built a PyTorch pipeline to tune GANs across multiple loss functions and optimizers; benchmarked Stable Diffusion vs. conditional DC-GANs on CelebA (IS=6.85, FID=164) and CIFAR-10 (IS=5.2, FID=92).

AutoBots: Reinforcement Learning for Autonomous Driving | *PyTorch, Reinforcement Learning, LiDAR*

- Designed and trained PPO and Actor-Critic agents for continuous vehicle control (steering, throttle, braking) using an 18D LiDAR-based state representation.
- Engineered reward shaping and 256-256-128 MLP actor/critic architectures; evaluated performance via cumulative reward, collision rate, and distance traveled.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, C, JavaScript, HTML, CSS, Git/GitHub, APIs

Software Tools: Pandas, Numpy, Matplotlib, Scikit-learn, Pytorch, Tensorflow, Apache Spark, AWS Cloud

Domains: Cloud, ML/AI, GenAI, Data Structures & Algorithms, Full-Stack, Data Visualization, Agile Development

Soft Skills: Project Management, Organization, Presentations, Detail-Oriented, Communication, Quick-Learner